Operating Systems – Optimal Page Replacement Algorithm

Practice Questions with Step-by-Step Solutions

1. Introduction

Optimal Page Replacement is a page replacement technique used in Operating Systems to minimize the number of page faults.

When a new page needs to be loaded and all page frames are full, the algorithm replaces the page that will not be used for the longest time in the future.

Important terms:

- **Page Hit:** The requested page is already available in main memory.
- **Page Miss (Page Fault):** The requested page is not available in main memory and must be loaded.
- **Page Frame:** A fixed-size storage location in main memory used to hold a page.

2. Formulas

Hit Ratio

Hit Ratio = (Total Hits / Total Page References) × 100

Miss Ratio

Miss Ratio = (Total Misses / Total Page References) × 100

Relationship:

Hit Ratio + Miss Ratio = 100%

Question 1

A system has 3 page frames and uses the Optimal Page Replacement algorithm. Initially, all frames are empty.

Reference String:

1, 2, 3, 4, 1, 2, 5, 1, 2, 3

Calculate:

1.

Total page hits.

2.
3.

Total page misses.

4.
5.

Hit ratio.

6.
7.

Miss ratio.

8.

Solution

Step 1: Initially, all frames are empty.

•

Page 1 is loaded into Frame 1 – Miss.

•
•

Page 2 is loaded into Frame 2 – Miss.

•
•

Page 3 is loaded into Frame 3 – Miss.

-

Step 2: Page 4 arrives, but all frames are full.

Current frames contain 1, 2, 3.

Future references: 4, 1, 2, 5, 1, 2, 3.

Page 3 is used farthest in the future, so replace page 3 with page 4.

Step 3: Pages 1 and 2 are already in memory, so both are Hits.

Step 4: Page 5 arrives.

Current frames contain 1, 2, 4.

Page 4 will never be used again, so replace page 4 with page 5.

Step 5: Pages 1 and 2 are Hits. Page 3 is not present, so it causes a Miss.

Page Replacement Table

Page	F1	F2	F3	Result
1	1	-	-	Miss
2	1	2	-	Miss
3	1	2	3	Miss
4	1	2	4	Miss
1	1	2	4	Hit
2	1	2	4	Hit
5	1	2	5	Miss
1	1	2	5	Hit
2	1	2	5	Hit
3	3	2	5	Miss

Calculation:

Total References = 10

Total Hits = 4

Total Misses = 6

Hit Ratio = (4 / 10) × 100 = **40%**

Miss Ratio = (6 / 10) × 100 = **60%**

Final Answer: Hit Ratio = 40%, Miss Ratio = 60%.

Question 2

A system uses 3 page frames with Optimal Page Replacement.

Reference String:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3

Calculate the total page faults, hit ratio and miss ratio.

Solution

Step 1: Load pages 7, 0 and 1 into the three empty frames. All three are Misses.

Step 2: Page 2 arrives.

Current frames: 7, 0, 1.

Future references: 2, 0, 3, 0, 4, 2, 3.

Page 7 is never used again. Replace page 7 with page 2.

Step 3: Page 0 is already present. It is a Hit.

Step 4: Page 3 arrives.

Current frames: 2, 0, 1.

Page 1 is never used again. Replace page 1 with page 3.

Step 5: Page 0 is a Hit.

Step 6: Page 4 arrives.

Current frames: 2, 0, 3.

Page 0 is never used again. Replace page 0 with page 4.

Step 7: Pages 2 and 3 are already in memory, so both are Hits.

Page Replacement Table

Page	F1	F2	F3	Result
7	7	-	-	Miss
0	7	0	-	Miss
1	7	0	1	Miss
2	2	0	1	Miss
0	2	0	1	Hit
3	2	0	3	Miss
0	2	0	3	Hit
4	2	4	3	Miss
2	2	4	3	Hit
3	2	4	3	Hit

Calculation:

Total References = 10

Total Hits = 4

Total Misses = 6

Hit Ratio = (4 / 10) × 100 = **40%**

Miss Ratio = (6 / 10) × 100 = **60%**

Final Answer: Hit Ratio = 40%, Miss Ratio = 60%.

Question 3

A computer system has 3 page frames and uses Optimal Page Replacement.

Reference String:

2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5

Calculate the total page faults and hit ratio.

Solution

Step 1: Load pages 2, 3 and 2.

- Page 2 – Miss.

-
-

Page 3 – Miss.

-
-

Page 2 – Hit.

-

Step 2: Page 1 arrives.

Load page 1 into the empty third frame. This is a Miss.

Step 3: Page 5 arrives.

Current frames: 2, 3, 1.

Page 1 is never used again. Replace page 1 with page 5.

Step 4: Page 2 is already present. It is a Hit.

Step 5: Page 4 arrives.

Current frames: 2, 3, 5.

Future references: 5, 3, 2, 5.

Page 2 will be used farthest in the future. Replace page 2 with page 4.

Step 6: Page 5 and page 3 are Hits.

Step 7: Page 2 arrives.

Current frames: 4, 3, 5.

Page 3 and page 4 will not be used again. Either can be replaced. Here, replace page 3 with page 2.

Step 8: Page 5 is already present. It is a Hit.

Page Replacement Table

Page	F1	F2	F3	Result
2	2	-	-	Miss
3	2	3	-	Miss
2	2	3	-	Hit

1	2	3	1	Miss
5	2	3	5	Miss
2	2	3	5	Hit
4	4	3	5	Miss
5	4	3	5	Hit
3	4	3	5	Hit
2	4	2	5	Miss
5	4	2	5	Hit

Calculation:

Total References = 11

Total Hits = 5

Total Misses = 6

Hit Ratio = (5 / 11) × 100 = **45.45%**

Miss Ratio = (6 / 11) × 100 = **54.55%**

Final Answer: Hit Ratio = 45.45%, Miss Ratio = 54.55%.

Question 4

A system has 4 page frames and uses the Optimal Page Replacement algorithm.

Reference String:

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Calculate the hit ratio and miss ratio.

Solution

Step 1: Load pages 1, 2, 3 and 4 into the four empty frames. All are Misses.

Step 2: Pages 1 and 2 are already available in memory. Both are Hits.

Step 3: Page 5 arrives.

Current frames: 1, 2, 3, 4.

Future references: 5, 1, 2, 3, 4, 5.

Page 4 will be used farthest in the future. Replace page 4 with page 5.

Step 4: Pages 1, 2 and 3 are already present. All are Hits.

Step 5: Page 4 arrives.

Current frames:

DEADLOCK

Additional Numerical Questions – Operating Systems

Resource Allocation and Deadlock

These questions are additional to the five Optimal Page Replacement problems in the previous document.

Question 5 – Resource Allocation (Banker's Algorithm)

A system has 5 processes (P0, P1, P2, P3, P4) and 3 types of resources (A, B, C).

The total resources available in the system are:

-

Resource A = 10

-
-

Resource B = 5

-
-

Resource C = 7

-

The current Allocation Matrix and Maximum Need Matrix are given below.

Allocation Matrix

Process	A	B	C
P0	0	1	0
P1	2	0	0
P2	3	0	2
P3	2	1	1
P4	0	0	2

Maximum Need Matrix

Process	A	B	C
P0	7	5	3
P1	3	2	2
P2	9	0	2
P3	2	2	2
P4	4	3	3

Calculate:

1.

Available resources.

2.
3.

Need Matrix.

4.
5.

Determine whether the system is in a safe state.

6.
7.

Find the safe sequence.

8.

Solution

Step 1: Calculate Available Resources

Formula:

Available = Total Resources − Allocated Resources

First, calculate the total allocated resources.

Resource	Total	Allocated	Available
A	10	7	3
B	5	2	3
C	7	5	2

Therefore:

Available = (3, 3, 2)

Step 2: Calculate Need Matrix

Formula:

Need = Maximum − Allocation

For example, for P0:

-

A = 7 − 0 = 7

-
-

B = 5 − 1 = 4

-
-

C = 3 − 0 = 3

-

Similarly, calculate for all processes.

Process	A	B	C
P0	7	4	3
P1	1	2	2
P2	6	0	0

P3 0 1 1

P4 4 3 1

Step 3: Check Safe State

Initially:

Available = (3, 3, 2)

A process can execute if:

Need ≤ Available

Check each process:

- P0: (7, 4, 3) – Cannot execute.
- P1: (1, 2, 2) – Can execute.
- P2: (6, 0, 0) – Cannot execute.
- P3: (0, 1, 1) – Can execute.
- P4: (4, 3, 1) – Cannot execute.

Select P1 first.

After P1 completes, its allocated resources are released.

New Available:

(3, 3, 2) + (2, 0, 0) = (5, 3, 2)

Now P3 can execute.

After P3 completes:

(5, 3, 2) + (2, 1, 1) = (7, 4, 3)

Now P4 can execute.

After P4 completes:

(7, 4, 3) + (0, 0, 2) = (7, 4, 5)

Now P0 can execute.

After P0 completes:

(7, 4, 5) + (0, 1, 0) = (7, 5, 5)

Finally, P2 can execute.

Safe Sequence Calculation

Step	Process	Available Before Execution	Allocated Resources Released	Available After Execution
1	P1	(3,3,2)	(2,0,0)	(5,3,2)
2	P3	(5,3,2)	(2,1,1)	(7,4,3)
3	P4	(7,4,3)	(0,0,2)	(7,4,5)
4	P0	(7,4,5)	(0,1,0)	(7,5,5)
5	P2	(7,5,5)	(3,0,2)	(10,5,7)

Final Answer:

Safe Sequence = P1 → P3 → P4 → P0 → P2

The system is in a **SAFE STATE** because all processes can complete successfully.

Question 6 – Deadlock Detection

A system has 3 processes (P0, P1, P2) and 2 types of resources (A and B).

The total resources in the system are:

- Resource A = 4

-
-

Resource B = 3

-

The current allocation and resource request matrices are given below.

Allocation Matrix

Process	A	B
P0	1	0
P1	1	1
P2	1	1

Request Matrix

Process	A	B
P0	0	1
P1	1	0
P2	1	1

Calculate:

1.

Available resources.

2.
3.

Determine whether any process can complete.

4.
5.

Identify whether the system is in deadlock.

6.
7.

If deadlock exists, identify the processes involved.

8.

Solution

Step 1: Calculate Available Resources

Formula:

Available = Total Resources − Allocated Resources

Resource	Total	Allocated	Available
A	4	3	1
B	3	2	1

Therefore:

Available = (1, 1)

Step 2: Check Process Requests

A process can complete if its current request can be satisfied using available resources.

Initially, Available = (1, 1).

Check each process:

Process	Request (A,B)	Available (A,B)	Can Execute?
P0	(0,1)	(1,1)	Yes
P1	(1,0)	(1,1)	Yes
P2	(1,1)	(1,1)	Yes

All three processes can execute initially. Therefore, this example does not demonstrate a deadlock yet.

To identify a deadlock, consider the following corrected request matrix instead.

Corrected Request Matrix

Process	A	B
P0	0	2
P1	2	0
P2	1	1

Step 3: Check Available Resources

Available = (1, 1)

Check the requests:

Process	Request	Available	Can Execute?
P0	(0,2)	(1,1)	No
P1	(2,0)	(1,1)	No
P2	(1,1)	(1,1)	Yes

P2 can execute.

After P2 completes, release its allocated resources (1,1).

New Available:

(1,1) + (1,1) = (2,2)

Now:

-

P0 requests (0,2), which can be satisfied.

-
-

P1 requests (2,0), which can also be satisfied.

-

Both processes can complete.

Therefore, the system is still not in deadlock.

Final Answer

The system is in a **SAFE STATE**, not a deadlock state.

A valid completion sequence is:

P2 → P0 → P1

Important: In deadlock detection, always check whether the outstanding request of a process can be satisfied with currently available resources. If it completes, release its allocated resources and check again.

Important Difference: Banker's Algorithm vs Deadlock Detection

Banker's Algorithm	Deadlock Detection
Prevents the system from entering an unsafe state.	Checks whether deadlock has already occurred.
Requires Maximum Need and Allocation matrices.	Uses Allocation and outstanding Request matrices.
Calculates Need = Maximum − Allocation.	Checks whether Request ⩽ Available.
Determines whether a safe sequence exists.	Determines which processes can finish and which remain blocked.
Used before granting resource requests.	Used to detect existing deadlocks.

Important Formulas

1.

Available = Total Resources − Allocated Resources

2.
3.

Need = Maximum − Allocation

4.
5.

Safe condition: Need ≤ Available

6.
7.

Deadlock detection condition: Request ≤ Available

8.

Remember: A safe state means there is at least one sequence in which all processes can finish. An unsafe state does not necessarily mean that a deadlock has already occurred.

Additional Numerical Questions – LRU Page Replacement Algorithm

1. Introduction

LRU stands for **Least Recently Used**.

It is a page replacement algorithm in which the page that has not been used for the longest time in the past is replaced when a new page arrives and all frames are full.

Important Rules:

1.

If the requested page is already in memory, it is a Hit.

2.
3.

If the requested page is not in memory, it is a Miss (Page Fault).

4.
5.

If an empty frame is available, load the new page into it.

6.
7.

If all frames are full, replace the page that was used least recently.

8.
9.

Unlike Optimal Page Replacement, LRU looks at past references, not future references.

10.

Formulas

Hit Ratio = (Total Hits / Total References) × 100

Miss Ratio = (Total Misses / Total References) × 100

Question 1 – Easy

A system has 3 page frames and uses the LRU Page Replacement algorithm. Initially, all frames are empty.

Reference String:

1, 2, 3, 1, 4, 5, 2, 1

Calculate:

1.

Total page hits.

2.
3.

Total page misses.

4.
5.

Hit ratio.

6.
7.

Miss ratio.

8.

Solution

Step 1: Pages 1, 2 and 3 arrive.

All frames are empty, so all three pages are loaded.

Step 2: Page 1 arrives again.

Page 1 is already present. It is a Hit.

Step 3: Page 4 arrives.

Current frames contain 1, 2, 3.

The recent usage history is:

-

Page 1: Used most recently.

-
-

Page 2: Used before page 1.

-
-

Page 3: Used least recently.

-

Therefore, replace page 3 with page 4.

Step 4: Page 5 arrives.

Current frames contain 1, 2, 4.

Page 2 was used least recently, so replace page 2 with page 5.

Step 5: Page 2 arrives.

Current frames contain 1, 5, 4.

Page 4 was used least recently, so replace page 4 with page 2.

Step 6: Page 1 arrives.

Page 1 is already present. It is a Hit.

Page Replacement Table

Page	F1	F2	F3	Result
1	1	-	-	Miss
2	1	2	-	Miss
3	1	2	3	Miss
1	1	2	3	Hit
4	1	2	4	Miss
5	1	5	4	Miss
2	1	5	2	Miss
1	1	5	2	Hit

Calculation:

Total References = 8

Total Hits = 2

Total Misses = 6

Hit Ratio = (2 / 8) × 100 = **25%**

Miss Ratio = (6 / 8) × 100 = **75%**

Final Answer: Hit Ratio = 25%, Miss Ratio = 75%.

Question 2 – Easy to Moderate

A computer system has 3 page frames and uses the LRU Page Replacement algorithm.

Reference String:

7, 0, 1, 2, 0, 3, 0, 4, 2, 3

Calculate the total page faults, hit ratio and miss ratio.

Solution

Step 1: Load pages 7, 0 and 1 into the empty frames. All are Misses.

Step 2: Page 2 arrives.

Current frames: 7, 0, 1.

Page 7 was used least recently. Replace 7 with 2.

Step 3: Page 0 arrives.

Page 0 is already in memory. Hit.

Step 4: Page 3 arrives.

Current frames: 2, 0, 1.

Page 1 was used least recently. Replace 1 with 3.

Step 5: Page 0 arrives again. Hit.

Step 6: Page 4 arrives.

Current frames: 2, 0, 3.

Page 2 was used least recently. Replace 2 with 4.

Step 7: Page 2 arrives.

Current frames: 4, 0, 3.

Page 3 was used least recently. Replace 3 with 2.

Step 8: Page 3 arrives.

Current frames: 4, 0, 2.

Page 0 was used least recently. Replace 0 with 3.

Page Replacement Table

Page	F1	F2	F3	Result
7	7	-	-	Miss
0	7	0	-	Miss
1	7	0	1	Miss
2	2	0	1	Miss
0	2	0	1	Hit
3	2	0	3	Miss
0	2	0	3	Hit
4	4	0	3	Miss
2	4	2	3	Miss
3	4	2	3	Hit

Calculation:

Total References = 10

Total Hits = 3

Total Misses = 7

Hit Ratio = (3 / 10) × 100 = **30%**

Miss Ratio = (7 / 10) × 100 = **70%**

Final Answer: Hit Ratio = 30%, Miss Ratio = 70%.

Question 3 – Moderate

A system has 3 page frames and uses LRU Page Replacement.

Reference String:

2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5

Calculate:

1.

Total page hits.

2.
3.

Total page faults.

4.
5.

Hit ratio.

6.
7.

Miss ratio.

8.

Solution

Step 1: Pages 2 and 3 arrive, followed by page 2 again.

-

Page 2 – Miss.

-
-

Page 3 – Miss.

-
-

Page 2 – Hit.

-

Step 2: Page 1 arrives.

Load page 1 into the empty frame. Miss.

Step 3: Page 5 arrives.

Current frames: 2, 3, 1.

Page 3 was used least recently. Replace page 3 with page 5.

Step 4: Page 2 arrives. Hit.

Step 5: Page 4 arrives.

Current frames: 2, 5, 1.

Page 1 was used least recently. Replace page 1 with page 4.

Step 6: Page 5 arrives. Hit.

Step 7: Page 3 arrives.

Current frames: 2, 5, 4.

Page 2 was used least recently. Replace page 2 with page 3.

Step 8: Page 2 arrives.

Current frames: 3, 5, 4.

Page 4 was used least recently. Replace page 4 with page 2.

Step 9: Page 5 is already present. Hit.

Page Replacement Table

Page	F1	F2	F3	Result
2	2	-	-	Miss
3	2	3	-	Miss
2	2	3	-	Hit
1	2	3	1	Miss
5	2	5	1	Miss
2	2	5	1	Hit
4	2	5	4	Miss

5	2	5	4	Hit
3	3	5	4	Miss
2	3	5	2	Miss
5	3	5	2	Hit

Calculation:

Total References = 11

Total Hits = 4

Total Misses = 7

Hit Ratio = (4 / 11) × 100 = **36.36%**

Miss Ratio = (7 / 11) × 100 = **63.64%**

Final Answer: Hit Ratio = 36.36%, Miss Ratio = 63.64%.

Question 4 – Moderate (4 Frames)

A system has 4 page frames and uses the LRU Page Replacement algorithm.

Reference String:

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Calculate the total page faults, hit ratio and miss ratio.

Solution

Step 1: Pages 1, 2, 3 and 4 are loaded into the four empty frames. All are Misses.

Step 2: Pages 1 and 2 arrive again. Both are Hits.

Step 3: Page 5 arrives.

Current frames: 1, 2, 3, 4.

Page 3 was used least recently. Replace page 3 with page 5.

Step 4: Pages 1 and 2 are already in memory. Both are Hits.

Step 5: Page 3 arrives.

Current frames: 1, 2, 5, 4.

Page 4 was used least recently. Replace page 4 with page 3.

Step 6: Page 4 arrives.

Current frames: 1, 2, 5, 3.

Page 5 was used least recently. Replace page 5 with page 4.

Step 7: Page 5 arrives.

Current frames: 1, 2, 4, 3.

Page 1 was used least recently. Replace page 1 with page 5.

Page Replacement Table

Page	F1	F2	F3	F4	Result
1	1	-	-	-	Miss
2	1	2	-	-	Miss
3	1	2	3	-	Miss
4	1	2	3	4	Miss
1	1	2	3	4	Hit
2	1	2	3	4	Hit
5	1	2	5	4	Miss
1	1	2	5	4	Hit
2	1	2	5	4	Hit
3	1	2	5	3	Miss
4	1	2	4	3	Miss
5	5	2	4	3	Miss

Calculation:

Total References = 12

Total Hits = 4

Total Misses = 8

Hit Ratio = (4 / 12) × 100 = **33.33%**

Miss Ratio = (8 / 12) × 100 = **66.67%**

Final Answer: Hit Ratio = 33.33%, Miss Ratio = 66.67%.

Question 5 – Difficult

A system has 3 page frames. Initially, all frames are empty. It uses LRU Page Replacement.

Reference String:

4, 7, 6, 1, 7, 6, 1, 2, 7, 2

Calculate:

1.

Total page hits.

2.
3.

Total page faults.

4.
5.

Hit ratio.

6.
7.

Miss ratio.

8.

Solution

Step 1: Pages 4, 7 and 6 arrive.

All frames are empty, so all three are Misses.

Step 2: Page 1 arrives.

Current frames: 4, 7, 6.

Page 4 was used least recently. Replace page 4 with page 1.

Step 3: Page 7 arrives. Hit.

Step 4: Page 6 arrives. Hit.

Step 5: Page 1 arrives. Hit.

Step 6: Page 2 arrives.

Current frames: 1, 7, 6.

Page 7 was used least recently. Replace page 7 with page 2.

Step 7: Page 7 arrives.

Current frames: 1, 2, 6.

Page 6 was used least recently. Replace page 6 with page 7.

Step 8: Page 2 arrives. Hit.

Page Replacement Table

Page	F1	F2	F3	Result
4	4	-	-	Miss
7	4	7	-	Miss
6	4	7	6	Miss
1	1	7	6	Miss
7	1	7	6	Hit
6	1	7	6	Hit
1	1	7	6	Hit
2	1	2	6	Miss
7	1	2	7	Miss
2	1	2	7	Hit

Calculation:

Total References = 10

Total Hits = 4

Total Misses = 6

Hit Ratio = (4 / 10) × 100 = **40%**

Miss Ratio = (6 / 10) × 100 = **60%**

Final Answer: Hit Ratio = 40%, Miss Ratio = 60%.

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6. Quick Answer Key

Question	Frames	References	Hits	Misses	Hit Ratio	Miss Ratio
1	3	8	2	6	25%	75%
2	3	10	3	7	30%	70%
3	3	11	4	7	36.36%	63.64%
4	4	12	4	8	33.33%	66.67%
5	3	10	4	6	40%	60%

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7. Difference Between FIFO, LRU and Optimal

Feature	FIFO	LRU	Optimal
Full Form	First In First Out	Least Recently Used	Optimal Page Replacement
Replacement Basis	Oldest loaded page	Least recently accessed page	Farthest future use
Looks At	Arrival time	Past references	Future references
Implementation	Queue	Timestamp/Stack	Future reference knowledge
Practical Usage	Simple implementation	Commonly used	Theoretical benchmark

Important Examination Tip

The most important difference between LRU and Optimal is:

-

LRU: Look backward and identify which page was used least recently.

-
-

Optimal: Look forward and identify which page will be used farthest in the future.

-

Always remember to update the recent usage of a page whenever it is accessed, including when a Hit occurs.

Additional Numerical Questions – CPU Scheduling Algorithms

1. Important Concepts and Formulas

Before solving CPU Scheduling numericals, understand these terms:

Term	Meaning
AT	Arrival Time – Time at which a process enters the ready queue.
BT	Burst Time – Total CPU execution time required by a process.
CT	Completion Time – Time at which a process finishes execution.
TAT	Turnaround Time – Total time taken from arrival to completion.
WT	Waiting Time – Total time a process waits in the ready queue.
RT	Response Time – Time from arrival until the process first gets CPU.

Important Formulas

Turnaround Time:

TAT = Completion Time − Arrival Time

Waiting Time:

WT = Turnaround Time − Burst Time

Response Time:

RT = First CPU Start Time − Arrival Time

Average Waiting Time:

Average WT = Total Waiting Time / Number of Processes

Average Turnaround Time:

Average TAT = Total Turnaround Time / Number of Processes

Question 1 – FCFS (First Come First Serve)

Consider the following processes with their Arrival Time and Burst Time.

Process	Arrival Time (AT)	Burst Time (BT)
P1	0	5
P2	1	3
P3	2	8
P4	3	6

Calculate:

1.

Completion Time (CT).

2.
3.

Turnaround Time (TAT).

4.
5.

Waiting Time (WT).

6.
7.

Average Waiting Time.

8.
9.

Average Turnaround Time.

10.

Solution

Step 1: Understand FCFS

FCFS executes processes in the order in which they arrive.

The process that arrives first gets the CPU first.

Arrival order:

P1 → P2 → P3 → P4

Step 2: Prepare the Gantt Chart

-

P1 executes from 0 to 5.

-
-

P2 executes from 5 to 8.

-
-

P3 executes from 8 to 16.

-
-

P4 executes from 16 to 22.

-

Gantt Chart

P1 P2 P3 P 4

0 5 8 16 22

Step 3: Calculate Completion Time

CT is the time at which a process finishes execution.

Process CT

P1 5

P2 8

P3 16

P4 22

Step 4: Calculate Turnaround Time

Formula:

TAT = CT − AT

Process CT AT TAT

P1 5 0 5

P2	8	1	7
P3	16	2	14
P4	22	3	19

Step 5: Calculate Waiting Time

Formula:

WT = TAT − BT

Process	TAT	BT	WT
P1	5	5	0
P2	7	3	4
P3	14	8	6
P4	19	6	13

Final Calculation

Total Waiting Time = 0 + 4 + 6 + 13 = 23

Average Waiting Time = 23 / 4 = **5.75**

Total Turnaround Time = 5 + 7 + 14 + 19 = 45

Average Turnaround Time = 45 / 4 = **11.25**

Final Answer:

-

Average Waiting Time = 5.75

-
-

Average Turnaround Time = 11.25

-

Question 2 – SJF (Shortest Job First – Non-Preemptive)

Consider the following processes:

Process	Arrival Time (AT)	Burst Time (BT)
P1	0	7
P2	2	4
P3	4	1
P4	5	4

Calculate the average waiting time and average turnaround time using Non-Preemptive SJF.

Solution

Step 1: Understand Non-Preemptive SJF

The process with the shortest burst time is selected from the available processes.

Once execution starts, it cannot be interrupted.

Step 2: Select Processes

At time 0:

Only P1 has arrived.

Therefore, P1 executes from 0 to 7.

At time 7:

Available processes are P2, P3 and P4.

Process	Burst Time
P2	4
P3	1
P4	4

P3 has the shortest burst time, so execute P3.

After P3, P2 and P4 both have burst time 4. We select P2 first because it arrived earlier.

Execution order:

P1 → P3 → P2 → P4

Step 3: Gantt Chart

P1 P3 P2 P

4

0 7 8 12 16

Step 4: Calculate CT, TAT and WT

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	7	7	7	0
P2	2	4	12	10	6
P3	4	1	8	4	3
P4	5	4	16	11	7

Final Calculation

Average Waiting Time:

(0 + 6 + 3 + 7) / 4 = **4**

Average Turnaround Time:

(7 + 10 + 4 + 11) / 4 = **8**

Final Answer:

- Average Waiting Time = 4
- Average Turnaround Time = 8

Question 3 – SRTF (Shortest Remaining Time First – Preemptive)

Consider the following processes:

Process	Arrival Time (AT)	Burst Time (BT)
P1	0	8
P2	1	4

P3	2	2
P4	3	1

Calculate the average waiting time and average turnaround time using SRTF.

Solution

Step 1: Understand SRTF

SRTF is the preemptive version of SJF.

The process having the shortest remaining execution time gets the CPU.

If a new process arrives with a shorter remaining time than the currently running process, the CPU switches to the new process.

Step 2: Process Execution

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Time 0: P1 starts execution.

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Time 1: P2 arrives. P1 has 7 units remaining, while P2 needs 4. P2 gets the CPU.

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Time 2: P3 arrives. P2 has 3 units remaining, while P3 needs 2. P3 gets the CPU.

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Time 3: P4 arrives. P3 and P4 both have 1 unit remaining. Continue P3.

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Time 4: P3 completes. P4 executes.

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-

Time 5: P4 completes. P2 resumes.

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Time 8: P2 completes. P1 resumes.

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Time 15: P1 completes.

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Step 3: Gantt Chart

P1 P2 P3 P4 P2 P 1

0 1 2 4 5 8 15

Step 4: Calculate Completion Time

Process	CT
P1	15
P2	8
P3	4
P4	5

Step 5: Calculate TAT and WT

Process	AT	BT	CT	TAT = CT − AT	WT = TAT − BT
P1	0	8	15	15	7
P2	1	4	8	7	3
P3	2	2	4	2	0
P4	3	1	5	2	1

Final Calculation

Average Waiting Time:

(7 + 3 + 0 + 1) / 4 = **2.75**

Average Turnaround Time:

(15 + 7 + 2 + 2) / 4 = **6.5**

Final Answer:

-

Average Waiting Time = 2.75

-
-

Average Turnaround Time = 6.5

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Question 4 – Round Robin Scheduling

Consider the following processes. All processes arrive at time 0.

Time Quantum = 2 milliseconds.

Process	Arrival Time (AT)	Burst Time (BT)
P1	0	5
P2	0	4
P3	0	2

Calculate:

1.

Completion Time.

2.
3.

Waiting Time.

4.
5.

Turnaround Time.

6.
7.

Average Waiting Time.

8.
9.

Average Turnaround Time.

10.

Solution

Step 1: Understand Round Robin

Round Robin is a preemptive scheduling algorithm.

Each process gets a fixed amount of CPU time called the Time Quantum.

If a process is not completed within its time quantum, it goes back to the ready queue.

Here:

Time Quantum = 2 ms.

Step 2: Process Execution

Initially, the ready queue is:

P1 → P2 → P3

Execution:

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P1 executes from 0 to 2. Remaining BT = 3.

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P2 executes from 2 to 4. Remaining BT = 2.

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P3 executes from 4 to 6. Completes.

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-

P1 executes from 6 to 8. Remaining BT = 1.

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-

P2 executes from 8 to 10. Completes.

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-

P1 executes from 10 to 11. Completes.

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Step 3: Gantt Chart

P1 P2 P3 P1 P2 P1

0 2 4 6 8 10 11

Step 4: Calculate Completion Time

Process	CT
P1	11
P2	10
P3	6

Step 5: Calculate TAT and WT

TAT = CT − AT

WT = TAT − BT

Process	AT	BT	CT	TAT	WT
P1	0	5	11	11	6
P2	0	4	10	10	6
P3	0	2	6	6	4

Final Calculation

Average Waiting Time:

(6 + 6 + 4) / 3 = **5.33 ms**

Average Turnaround Time:

(11 + 10 + 6) / 3 = **9 ms**

Final Answer:

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Average Waiting Time = 5.33 ms

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Average Turnaround Time = 9 ms

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Question 5 – Priority Scheduling (Non-Preemptive)

Consider the following processes:

Process	Arrival Time	Burst Time	Priority
P1	0	4	3
P2	0	3	1
P3	0	5	4
P4	0	2	2

Assume that a **smaller priority number indicates higher priority**.

Calculate the average waiting time and average turnaround time.

Solution

Step 1: Understand Priority Scheduling

In Priority Scheduling, the process with the highest priority gets the CPU first.

Here, the smallest priority number represents the highest priority.

Step 2: Arrange Processes According to Priority

Process	Burst Time	Priority
P2	3	1
P4	2	2
P1	4	3
P3	5	4

Execution order:

P2 → P4 → P1 → P3

Step 3: Gantt Chart

P2 P4 P1 P3

0 3 5 9 14

Step 4: Calculate CT, TAT and WT

Process	AT	BT	Priority	CT	TAT	WT
P1	0	4	3	9	9	5
P2	0	3	1	3	3	0
P3	0	5	4	14	14	9
P4	0	2	2	5	5	3

Final Calculation

Average Waiting Time:

(5 + 0 + 9 + 3) / 4 = **4.25**

Average Turnaround Time:

(9 + 3 + 14 + 5) / 4 = **7.75**

Final Answer:

- Average Waiting Time = 4.25
- Average Turnaround Time = 7.75

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6. Quick Answer Key

Question	Algorithm	Average WT	Average TAT
1	FCFS	5.75	11.25
2	SJF Non-Preemptive	4	8
3	SRTF Preemptive	2.75	6.5
4	Round Robin (Q = 2)	5.33	9

5	Priority Non-Preemptive	4.25	7.75